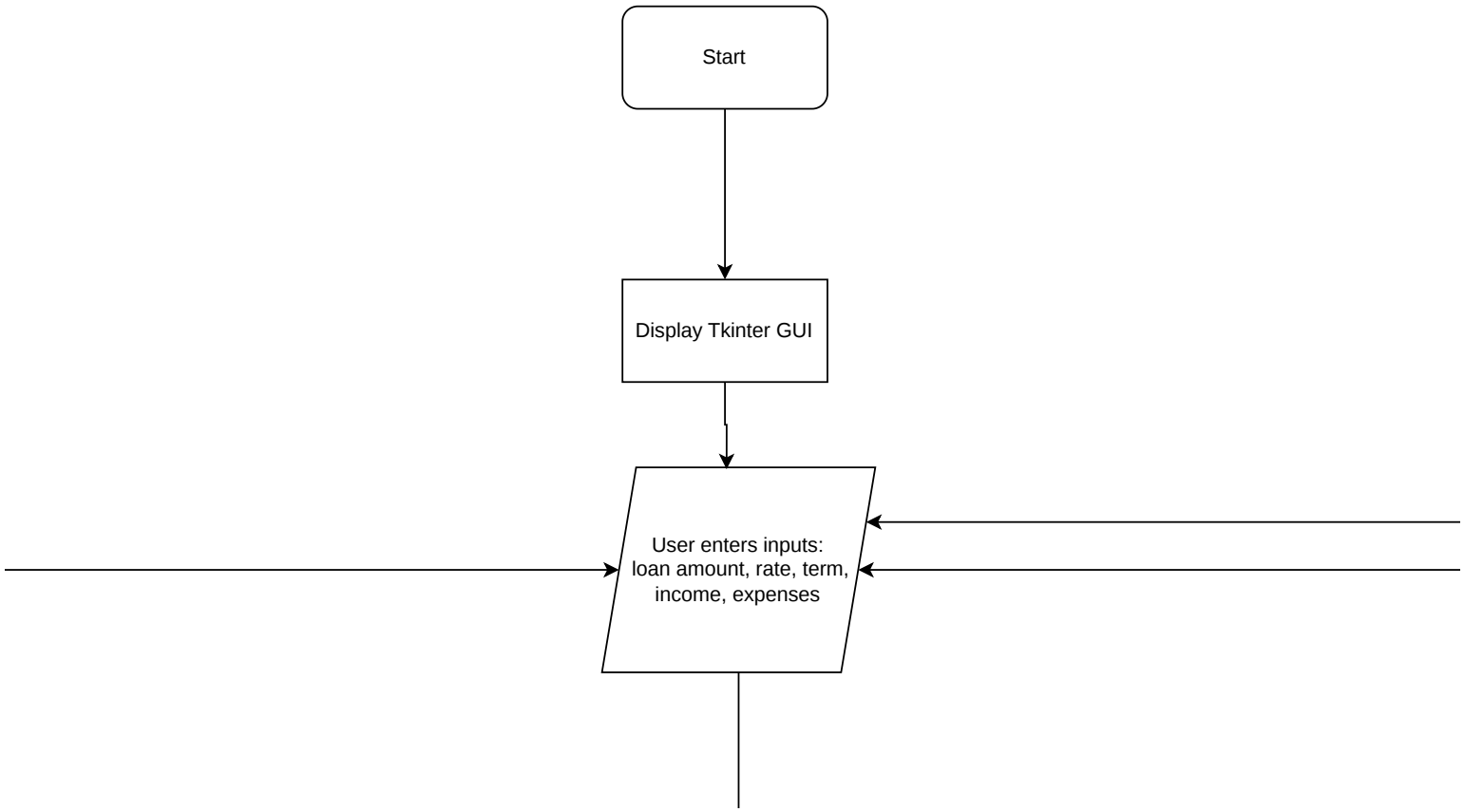
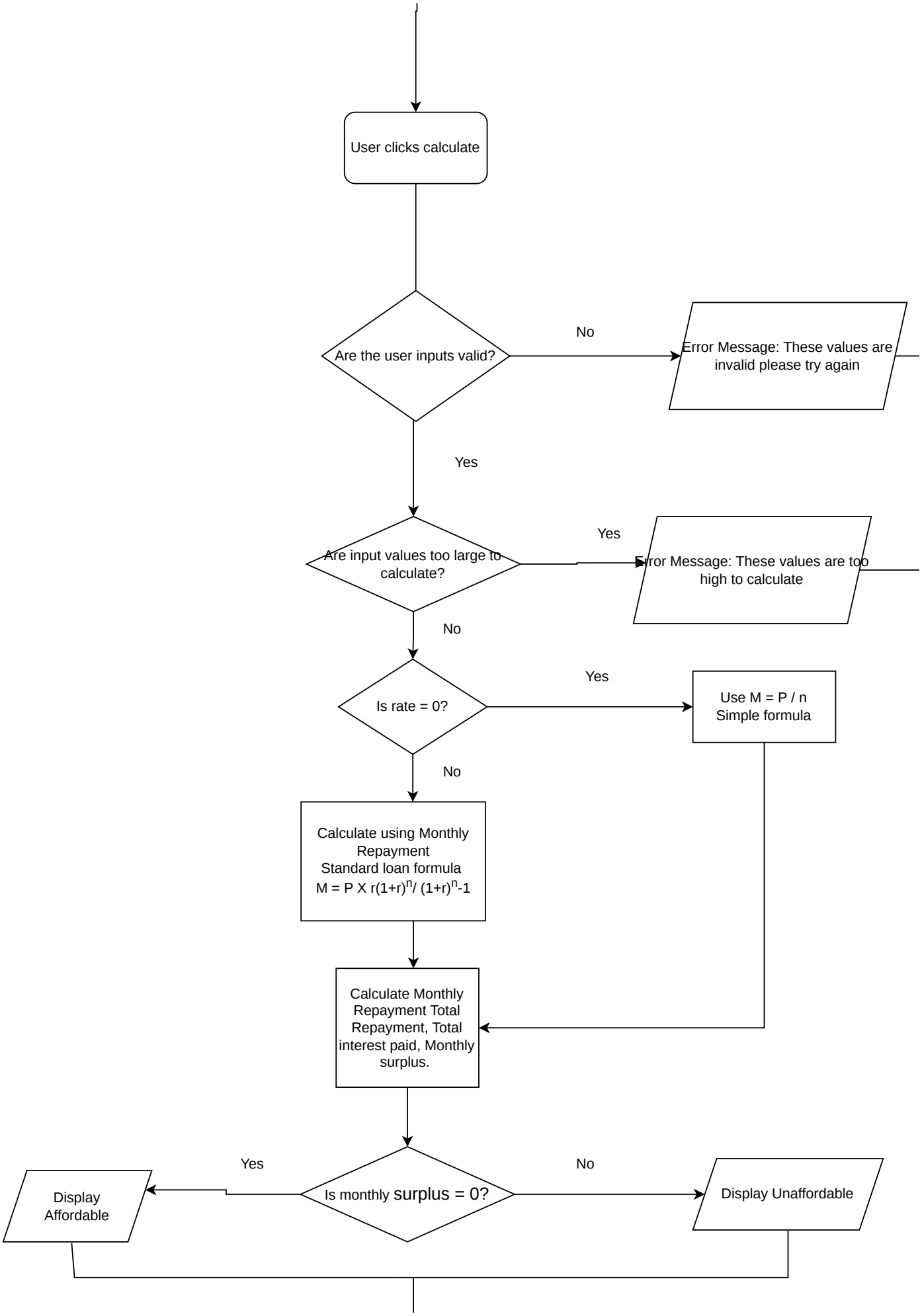
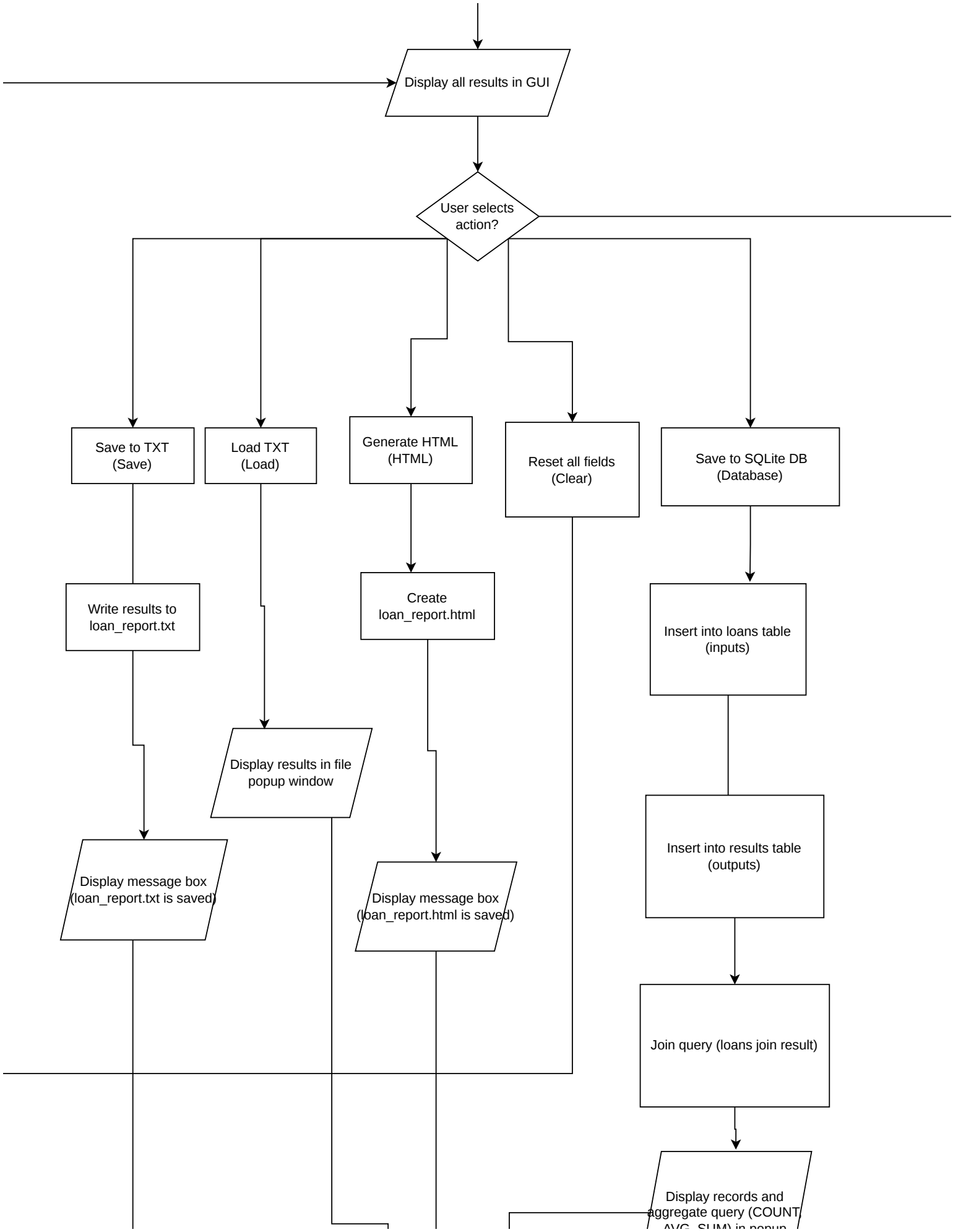
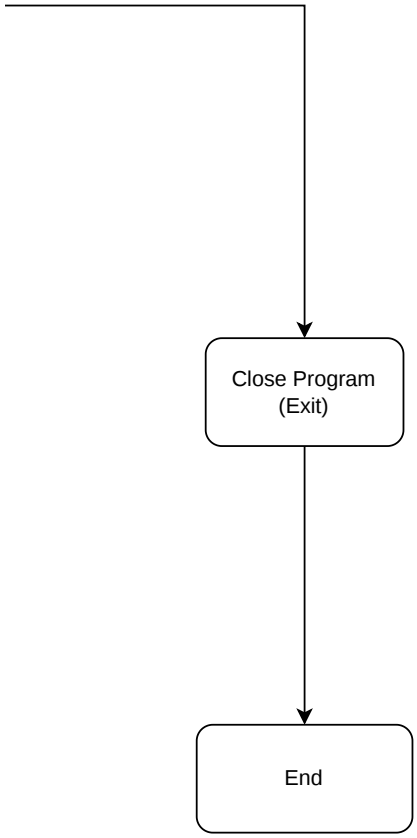
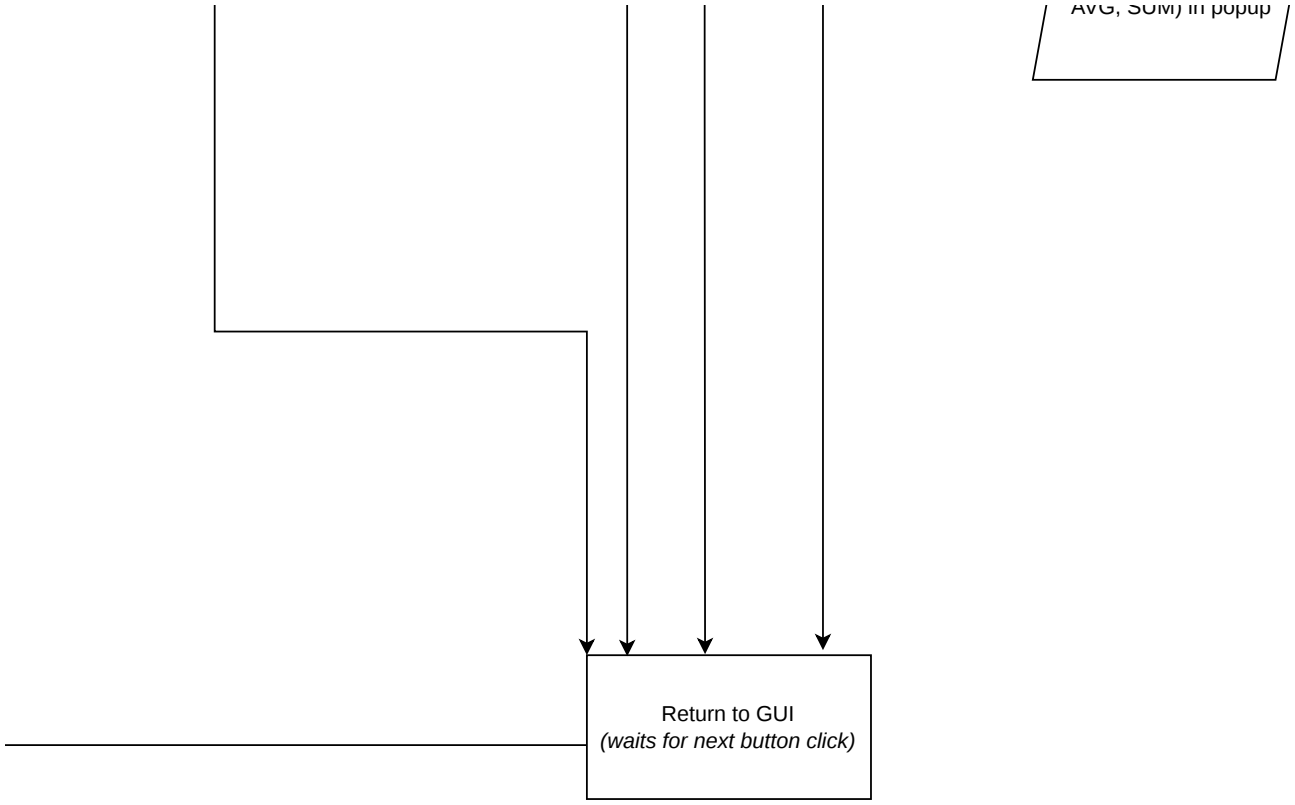

Start
Display Tkinter GUI
User enters inputs:
loan amount, rate, term,
income, expenses


User clicks calculate
Are the user inputs valid?
No
Error Message: These values are invalid please try again
Yes
Are input values too large to calculate?
Yes
Error Message: These values are too high to calculate
No
Is rate = 0?
Yes
Use M = P / n
Simple formula
No
Calculate using Monthly Repayment
Standard loan formula
M = P X r(1+r)^n/ (1+r)^n-1
Calculate Monthly Repayment Total Repayment, Total interest paid, Monthly surplus.
Is monthly surplus = 0?
Yes
Display Affordable
No
Display Unaffordable


Display all results in GUI
User selects action?
Save to TXT (Save)
Load TXT (Load)
Generate HTML (HTML)
Reset all fields (Clear)
Save to SQLite DB (Database)
Write results to loan_report.txt
Create loan_report.html
Insert into loans table (inputs)
Display results in file popup window
Insert into results table (outputs)
Display message box (loan_report.txt is saved)
Display message box (loan_report.html is saved)
Join query (loans join result)
Display records and aggregate query (COUNT, AVG, SUM) in popup


Close Program
(Exit)
End


AVG, SUM) in popup
Return to GUI
(waits for next button click)